

Bank Customer Retention Intelligence Dashboard:

A Machine Learning Approach to Predicting and Preventing Customer Churn

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Abstract

Customer churn prediction has emerged as a critical area in banking analytics, as retaining existing customers is substantially more cost-effective than acquiring new ones. This paper presents the design, development, and evaluation of a Bank Customer Retention Intelligence Dashboard—an end-to-end data science solution that combines exploratory data analysis, machine learning-based churn prediction, and an interactive Streamlit web application. The system was developed and deployed as part of an internship at Unified Mentor. Using a European bank dataset containing demographic, financial, and behavioral attributes of 10,000 customers, we built and evaluated multiple classification models including Logistic Regression, Random Forest, and Gradient Boosting. Key features influencing churn—such as age, balance, number of products, geographic location, and active membership status—were identified and visualized. The deployed dashboard enables business analysts to interactively explore churn risk segments, product utilization patterns, and retention strategy recommendations without requiring programming knowledge. Experimental results demonstrate that ensemble tree-based models achieve strong predictive accuracy, making this solution practically viable for real-world banking environments.

Keywords: Customer Churn, Machine Learning, Banking Analytics, Retention Strategy, Streamlit Dashboard, Random Forest, Customer Segmentation

1. Introduction

The banking and financial services industry faces intense competition and evolving customer expectations. In this landscape, customer retention has become a strategic priority, as studies consistently show that acquiring a new customer costs five to seven times more than retaining an existing one. Customer churn—when a client voluntarily closes their account or stops using a bank's services—directly impacts revenue, brand reputation, and long-term business viability.

Traditional approaches to understanding churn have relied on retrospective reporting and simple demographic analysis. However, the proliferation of digital banking data and advances in machine learning provide an opportunity to build predictive systems that can identify at-risk customers before they leave. Such proactive systems enable banks to deploy targeted retention strategies, including personalized offers, improved service outreach, and product bundling.

This paper presents the Bank Customer Retention Intelligence Dashboard, a comprehensive data science project developed during an internship at Unified Mentor. The project encompasses the complete data science lifecycle: data ingestion and preprocessing, exploratory data analysis (EDA), feature engineering, machine learning model development, and deployment as an interactive web dashboard. The dataset used is a European bank customer dataset with demographic and transactional features.

The key contributions of this work are:

- A thorough EDA revealing patterns in customer demographics, product usage, and churn behavior.
- Development and comparison of multiple machine learning classifiers for churn prediction.
- Feature importance analysis identifying the primary drivers of customer attrition.
- An interactive Streamlit dashboard enabling non-technical stakeholders to explore insights and apply the model in real time.
- Actionable retention strategy recommendations derived from data insights.

2. Literature Review

Customer churn prediction in banking has been studied extensively. Vafeiadis et al. (2015) evaluated multiple machine learning algorithms including Support Vector Machines, k-Nearest Neighbors, and Naive Bayes for bank churn prediction, demonstrating that ensemble methods outperform single classifiers. Xie et al. (2009) proposed an improved boosted decision tree approach, achieving high recall for identifying churners in imbalanced datasets.

Feature selection has been shown to play a crucial role in churn model performance. Researchers have consistently found that account balance, product count, customer age, and active membership status are among the strongest predictors of churn. Geographic segmentation has also emerged as a key factor in European banking studies, with customers from different regions exhibiting significantly different churn propensities.

Interactive visualization dashboards have increasingly been adopted in financial analytics to bridge the gap between data science outputs and business decision-making. Tools such as Streamlit, Dash, and Power BI enable data professionals to build deployable, user-friendly interfaces without extensive front-end development expertise. This project builds on this trend by integrating a production-grade Streamlit application with an underlying machine learning pipeline.

3. Dataset Description

The dataset used in this study is the European Bank Customer dataset (European_Bank.csv), which contains 10,000 records of bank customers with 14 feature columns. The dataset was sourced from a publicly available repository commonly used for churn prediction benchmarking.

3.1 Dataset Features

Feature	Type	Description
CreditScore	Numerical	Customer's credit score
Geography	Categorical	Country (France, Germany, Spain)
Gender	Categorical	Male / Female
Age	Numerical	Customer age in years
Tenure	Numerical	Years with the bank
Balance	Numerical	Account balance
NumOfProducts	Numerical	Number of bank products used
HasCrCard	Binary	Whether customer has a credit card
IsActiveMember	Binary	Whether customer is an active member
EstimatedSalary	Numerical	Estimated annual salary
Exited	Binary	Churn label: 1 = churned, 0 = retained

3.2 Class Distribution

The dataset exhibits a class imbalance with approximately 20.4% of customers labeled as churned (Exited = 1) and 79.6% as retained (Exited = 0). This imbalance was addressed during model training through class weighting and evaluation using metrics beyond simple accuracy.

4. Methodology

4.1 Data Preprocessing

Raw data was loaded and examined for missing values, duplicate records, and data type inconsistencies. Categorical features (Geography, Gender) were encoded using one-hot encoding to produce numeric representations suitable for machine learning algorithms. Numerical features were standardized using z-score normalization to ensure that distance-based and gradient-based models were not biased by differing feature scales.

4.2 Exploratory Data Analysis

Comprehensive EDA was conducted to uncover patterns and relationships within the data. Key findings included:

- Customers from Germany exhibited a significantly higher churn rate (~32%) compared to France (~16%) and Spain (~17%).
- Female customers churned at a higher rate (~25%) than male customers (~16%).
- Customers with only one or no bank products showed higher churn rates; those with four products had near-100% churn, suggesting over-commitment may cause dissatisfaction.
- Account balance showed a bimodal distribution; customers with zero balance had higher churn, as did those in very high balance brackets.
- Inactive members churned at nearly twice the rate of active members.

4.3 Feature Engineering

Several derived features were created to enrich the predictive signal:

- Balance-to-Salary Ratio: Relative financial standing of the customer.
- Age Group: Bucketed age ranges to capture lifecycle effects.
- Product Engagement Score: A composite score reflecting product utilization depth.
- Churn Risk Score: A weighted composite of key predictors used for segmentation visualization.

4.4 Machine Learning Models

Three classification algorithms were trained and evaluated:

- Logistic Regression: A baseline linear model providing interpretable coefficients.
- Random Forest Classifier: An ensemble of decision trees offering robust performance and feature importance scores.
- Gradient Boosting Classifier: A sequential ensemble method that iteratively corrects prediction errors.

Models were trained using a 80/20 train-test split with stratification to maintain class distribution. Hyperparameter tuning was performed using 5-fold cross-validation on the training set.

4.5 Evaluation Metrics

Given the class imbalance, accuracy alone is insufficient. Models were evaluated using Precision, Recall, F1-Score, and AUC-ROC (Area Under the Receiver Operating Characteristic Curve), which provides a comprehensive view of classifier discrimination ability.

5. Results and Discussion

5.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	AUC-ROC
Logistic Regression	80.3%	57.2%	19.8%	0.762
Random Forest	86.7%	74.1%	47.3%	0.856
Gradient Boosting	87.1%	73.8%	51.2%	0.869

Gradient Boosting achieved the best overall AUC-ROC (0.869), indicating strong discriminative ability. Random Forest provided a close alternative with slightly better precision. Logistic Regression performed adequately on accuracy but exhibited poor recall due to the class imbalance issue and the linear decision boundary's limitation with this dataset.

5.2 Feature Importance

Feature importance analysis from the Random Forest model revealed the following ranking of key churn drivers:

- Age — the strongest individual predictor of churn; older customers churned more frequently.
- Balance — customers with zero or very high balances were significantly more likely to churn.
- Number of Products — product count showed a non-linear relationship with churn.
- IsActiveMember — inactive members were nearly twice as likely to churn.
- Geography (Germany) — German customers exhibited elevated churn propensity.
- CreditScore — moderate negative correlation with churn.

5.3 Customer Segmentation

Customers were segmented into four churn risk tiers—Low, Medium, High, and Critical—using a composite risk scoring methodology. This segmentation enables the bank to allocate retention resources proportionally, focusing highest effort on Critical-risk customers who account for a disproportionate share of expected attrition.

6. Interactive Streamlit Dashboard

The Streamlit dashboard (deployed at <https://bank-customer-retention-dashboard.streamlit.app/>) provides business users with an interactive interface to explore model outputs without requiring programming expertise. The dashboard comprises five main sections:

- Overview & KPIs: High-level metrics including overall churn rate, total customers, average balance, and risk distribution.
- Customer Behavior Analysis: Visualizations of churn patterns across demographic groups, geographies, and membership status.
- Product Utilization Insights: Analysis of how product engagement relates to retention and revenue contribution.
- Churn Risk Segmentation: Interactive scatter and bar charts displaying customer risk scores with filtering capabilities.

- Individual Prediction Tool: A form where analysts can input a customer's attributes and receive a real-time churn probability estimate from the trained model.

The dashboard was built using Python's Streamlit library with Plotly for interactive visualizations. The trained machine learning model is serialized and loaded at runtime, enabling instant predictions without retraining.

7. Data-Driven Retention Strategies

Based on the analytical findings, the following targeted retention strategies are recommended:

7.1 High-Balance Inactive Customers

Customers with high account balances who are classified as inactive present a high churn risk despite their financial engagement. Banks should deploy relationship managers to personally re-engage these customers with customized investment or wealth management offerings.

7.2 German Market Segment

The elevated churn rate among German customers suggests localized dissatisfaction or stronger competitive alternatives. A dedicated regional customer experience improvement initiative—including local language service enhancements and Germany-specific product offerings—is recommended.

7.3 Older Customer Segment

Age emerged as the top churn predictor. Retirement-age customers may be switching to institutions offering better pension or savings products. Tailored senior banking packages and simplified digital interfaces may improve retention in this cohort.

7.4 Single-Product Customers

Customers using only one bank product show elevated churn vulnerability. Cross-selling relevant products—such as credit cards to savings account holders—can increase switching costs and deepen the customer-bank relationship.

8. Internship at Unified Mentor

This project was developed as part of a Data Science internship at Unified Mentor, an organization committed to nurturing emerging technology talent through practical, industry-relevant project experiences. The internship provided a structured environment to apply machine learning and data visualization skills to a real-world business problem.

Throughout the internship, the author independently handled all phases of the project: data acquisition and cleaning, exploratory analysis, model development, dashboard construction, and public deployment. Mentorship and code reviews conducted by Unified Mentor's technical team ensured adherence to industry best practices in documentation, version control (GitHub), and reproducible research.

The experience reinforced the importance of framing data science work within a business context—ensuring that technical outputs translate into actionable insights for non-technical stakeholders, as embodied by the interactive dashboard deliverable.

9. Conclusion

This paper presented a complete bank customer churn prediction and retention intelligence system developed during an internship at Unified Mentor. Using a European bank dataset, we demonstrated that ensemble machine learning methods—particularly Gradient Boosting—can effectively identify at-risk customers with AUC-ROC scores approaching 0.87. The companion Streamlit dashboard bridges the gap between model predictions and business decision-making by providing an intuitive, interactive interface for exploration and real-time prediction.

Key findings establish age, account balance, product count, and active membership status as the primary drivers of customer attrition. Geography-specific interventions and cross-selling strategies were identified as high-impact retention levers. Future work may explore deep learning architectures, real-time streaming data integration, and reinforcement learning-based personalization of retention offers.

This project demonstrates that data science can be a powerful enabler of customer-centric banking strategy when it is paired with interpretable outputs and accessible visualization tools.

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Project Repository: github.com/Prasannasegabandi36/bank-customer-retention-dashboard